

Machine Translation system for WIPO goods and services

ปวีณกร ไสรัตน์
รหัส 500307
บ้าน EXP

SEASON

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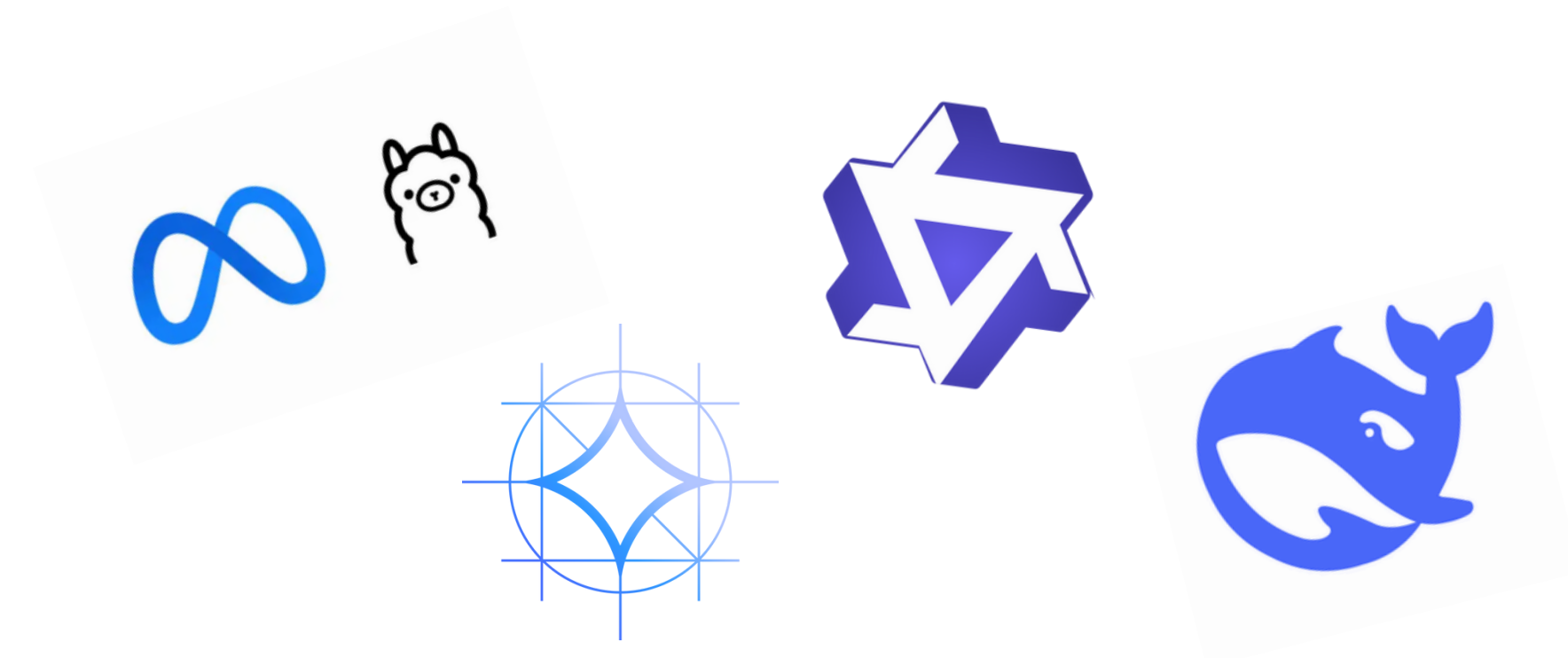
SUPER AI ENGINEER

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Overview

- Translation from English to Thai
- text pair with WIPO category
- Expectation: modern and reproduce



Resources

- 1.verified train (37k) and test dataset (8.4k records)
- 2.raw dataset (~1.2M records)
- 3.lanta GPU Partition

On Creating an English-Thai Code-switched Machine Translation in Medical Domain

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Abstract

Machine translation (MT) in the medical domain plays a pivotal role in enhancing health-care quality and disseminating medical knowledge. Despite advancements in English-Thai MT technology, common MT approaches often underperform in the medical field due to their inability to precisely translate medical terminologies. Our research prioritizes not merely improving translation accuracy but also maintaining medical terminology in English within the translated text through code-switched (CS) translation. We developed a method to produce CS medical translation data, fine-tuned a CS translation model with this data, and evaluated its performance against strong baselines, such as Google Neural Machine Translation (NMT) and GPT-3.5/GPT-4. Our model demonstrated competitive performance in automatic metrics and was highly favored in human preference evaluations. Our evaluation result also shows that medical professionals significantly prefer CS translations that maintain critical English terms accurately, even if it slightly compromises fluency. Our code and test set are publicly available https://github.com/preceptorai-org/NLLB_CS_EM_NLP2024.

1 Introduction

Medical-domain machine translation (MT) serves as a critical component in enhancing healthcare quality and disseminating medical knowledge. By providing accurate translations of medical research publications, MT enables local medical professionals without English proficiency to overcome the linguistic barrier and have access to more medical academic resources, which are predominantly written in English (Pecina et al., 2014; McLean et al., 2013). This accessibility is crucial for facilitating continuing medical education, which has been

Source Lang:	1. Hyperinflation (air trapping): In allergic asthma, the immune system overreacts to an allergen...
Target Lang:	1.ภาวะเงินเฟ้อทาง (การกักอากาศ): ในโรคหอบหืดจากภูมิแพ้ ระบบภูมิคุ้มกันจะตอบสนองต่อสารก่อภูมิแพ้มากเกินไป...
MD preferred:	1. Hyperinflation (air trapping): ใน allergic asthma ระบบภูมิคุ้มกันตอบสนองเกินต่อ allergen...

Figure 1: Example of how Google NMT alters the meaning of the sentence when translating from the source language (English) to the target language (Thai) and how it compares with the translations that Medical Doctors (MDs) prefer. “Hyperinflation” (abnormal increase of lung volume) is translated into Hyperinflation in economic context; “air trapping” (retention of air in the lungs distal to an obstruction) is translated into “air quarantine”.

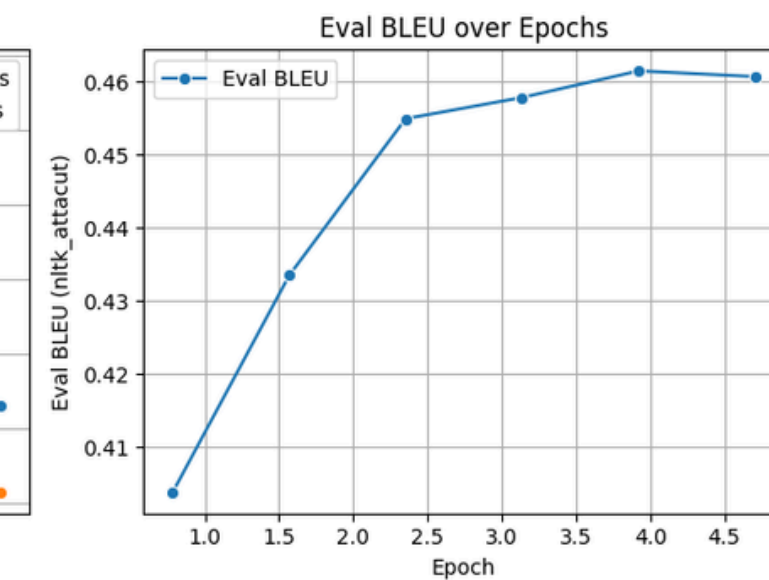
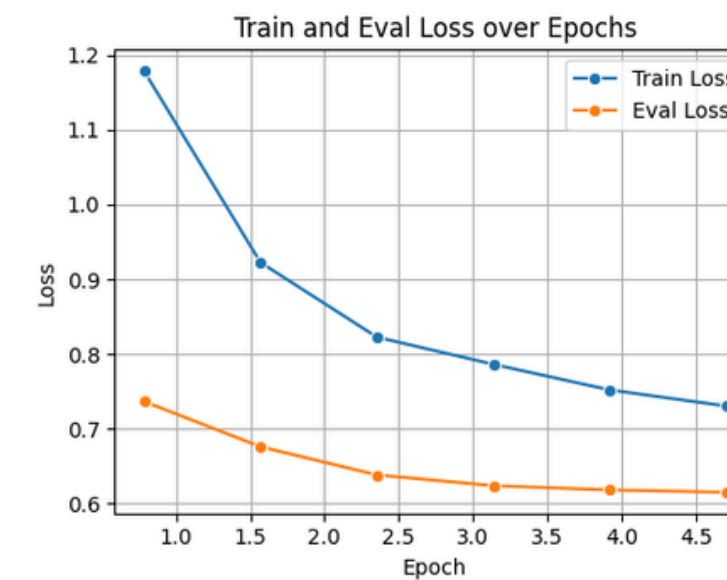
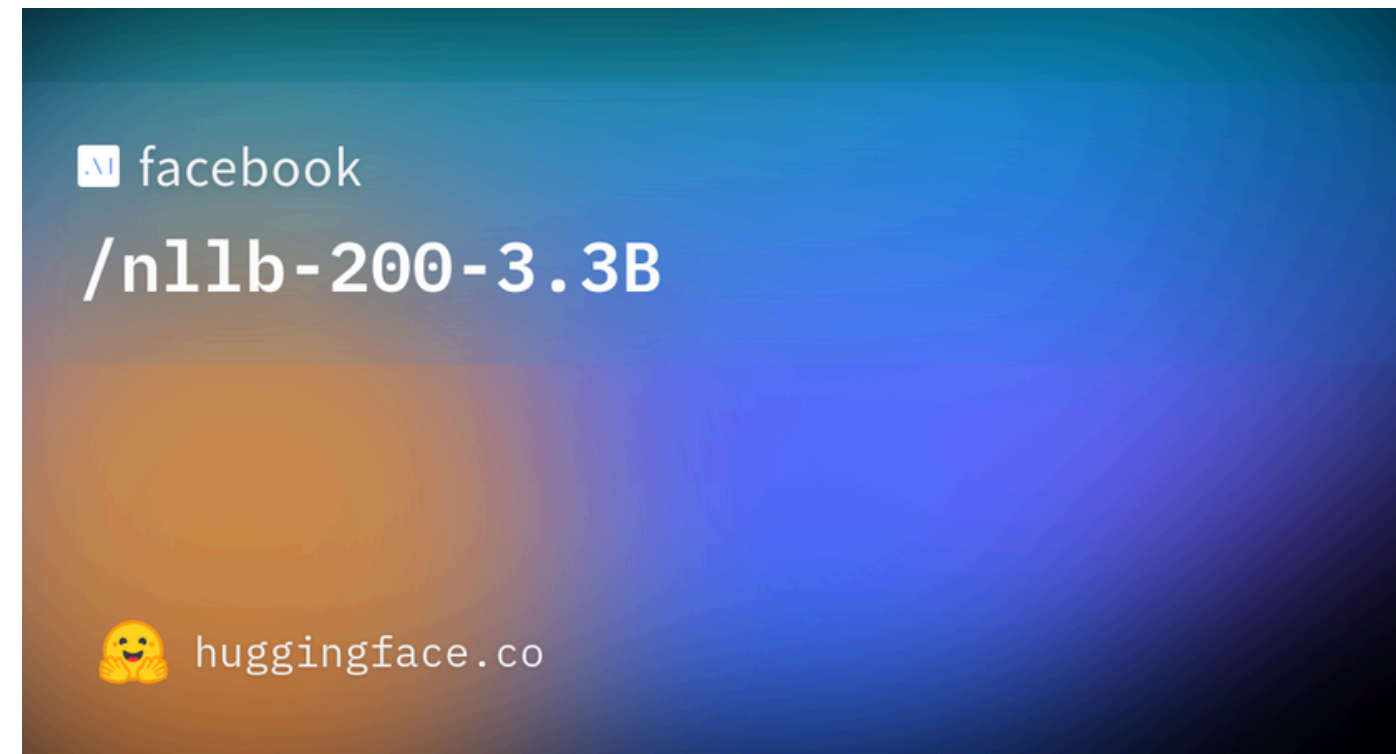
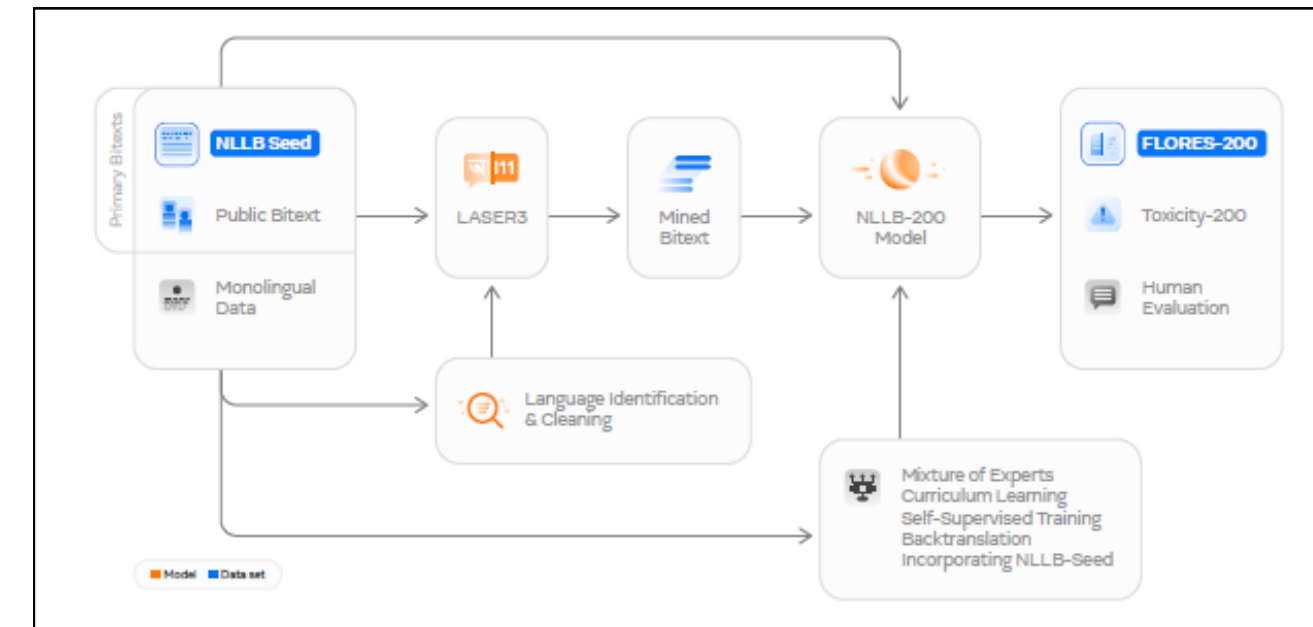
shown to be an effective strategy for healthcare professionals to enhance care quality and patient outcomes (Bloom, 2005; Randhawa et al., 2013).

Despite the research in the English-Thai MT field, most of the common MT techniques are not yet suitable for the medical domain due to the need for precise translation of medical terminology. Translating medical terminology accurately is challenging due to the lack of equivalent Thai terms for some English medical keywords. Thus, it is understandable why common techniques of machine translation in the medical domain MT — such as Google NMT, No Language Left Behind (NLLB) (Team et al., 2022), GPT-4 (Achiam et al., 2023), or Gemini-Pro (Team et al., 2023) — cannot translate medical keywords precisely, as shown in Figure 1. Previous studies have aimed to improve translation accuracy through terminology integration (Nieminen, 2023; Semenov et al., 2023), yet their application to the Thai language and specifi-

Model Variant	CS F1	BLEU	chrF	CER	WER	COMET	METEOR	Fact.
Gemini-Pro-CS	0.132	0.353	0.595	0.599	0.686	0.849	0.622	5.750
Gemini-Pro-MN	0.110	0.352	0.599	0.426	0.526	0.854	0.630	5.375
Google-NMT	0.119	0.385	0.617	0.392	0.480	0.815	0.650	3.125
GPT-3.5-CS	0.141	0.208	0.504	0.833	0.987	0.671	0.491	3.625
GPT-3.5-MN	0.114	0.205	0.504	0.599	0.775	0.687	0.494	3.875
GPT-4-CS	0.340	0.314	0.601	0.636	0.757	0.850	0.593	6.250
GPT-4-MN	0.132	0.282	0.581	0.511	0.660	0.847	0.597	5.125
Llama2-13B-CS	0.058	0.012	0.189	5.226	6.104	0.153	0.129	1.125
Llama2-13B-MN	0.082	0.022	0.224	3.326	4.139	0.163	0.174	1.000
Llama2-7B-CS	0.074	0.012	0.171	5.361	6.141	0.159	0.117	0.500
Llama2-7B-MN	0.086	0.015	0.197	3.852	4.761	0.162	0.143	0.500
OpenThaiGPT-13B-CS	0.039	0.094	0.446	2.343	2.538	0.394	0.388	2.125
OpenThaiGPT-13B-MN	0.036	0.094	0.465	1.978	2.208	0.425	0.396	2.375
OpenThaiGPT-7B-CS	0.045	0.046	0.308	12.620	13.224	0.310	0.237	2.875
OpenThaiGPT-7B-MN	0.027	0.068	0.344	9.954	10.223	0.369	0.282	2.750
SeaLLM-7B-CS	0.035	0.017	0.242	11.678	11.165	0.235	0.188	2.000
SeaLLM-7B-MN	0.076	0.032	0.329	8.340	8.259	0.321	0.259	1.705
Typhoon-7B-CS	0.021	0.012	0.220	18.946	18.434	0.186	0.168	1.875
Typhoon-7B-MN	0.023	0.013	0.239	18.111	19.020	0.174	0.176	1.875
NLLB	0.107	0.140	0.432	0.610	0.906	0.530	0.405	2.500
NLLB-1	0.475	0.253	0.487	0.491	0.593	0.678	0.502	4.375
NLLB-2	0.230	0.262	0.548	0.448	0.612	0.720	0.546	3.375
NLLB-3	0.380	0.257	0.520	0.472	0.604	0.702	0.521	4.000
NLLB-4	0.452	0.272	0.520	0.461	0.577	0.710	0.532	3.875
NLLB-5	0.193	0.255	0.544	0.458	0.627	0.715	0.546	3.250
NLLB-6	0.286	0.264	0.539	0.456	0.606	0.711	0.541	4.000

Baseline

- NLLB fine-tuning
- QLoRA rank 32



Proof of Concept (PoC)

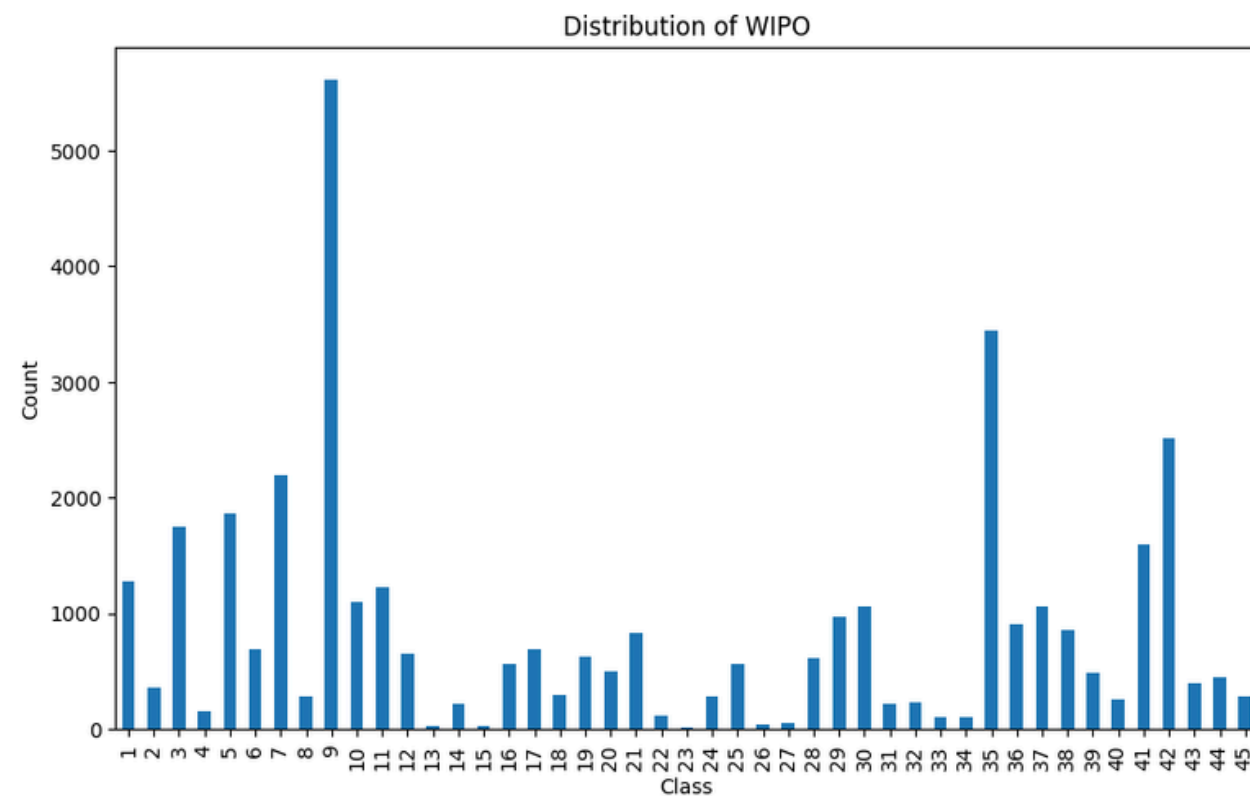
Model	CER	BLEU
Typhoon-translate	0.97	0.25
Typhoon-translate (FT)	0.59	0.28
SeedX-7B (FT)	0.62	0.20
Qwen3-8B	0.65	0.21
Qwen3-8B (FT)	0.62	0.23
Qwen3-14B	0.62	0.19
Qwen3-14B (FT)	0.68	0.28
Gemma3-12B-it	0.55	0.21
Gemma3-12B-it (FT)	0.51	0.23

**Performance didn't met
transformer-based at all**

Data Exploration

Data Exploration

- verified train dataset



class imbalance

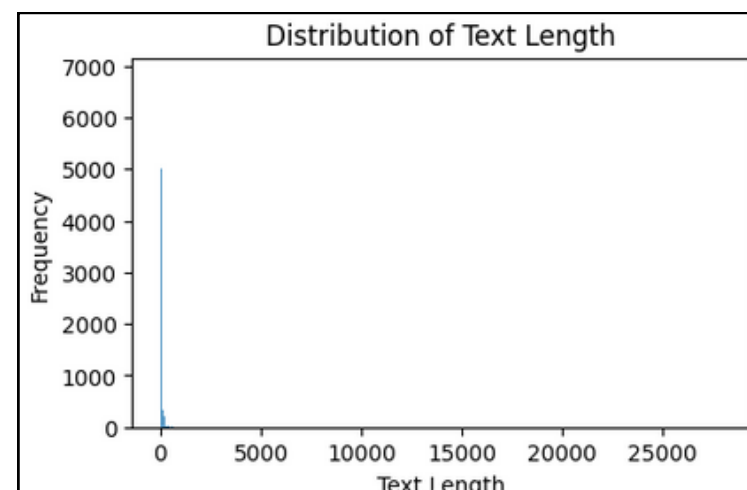
**Some class contains
a few sample**

Data Exploration

- raw dataset (~1.2M records)
- filter only unique pair of ENG and THA (~291k)

ENG	THA	COUNT
business administration	[การบริหารธุรกิจ, บริหารธุรกิจ]	2
business management	[การจัดการด้านธุรกิจ, การจัดการธุรกิจ, จัดการธ...	3
employment agencies	[ตัวแทนจัดหางาน, บริษัทจัดหางาน]	2

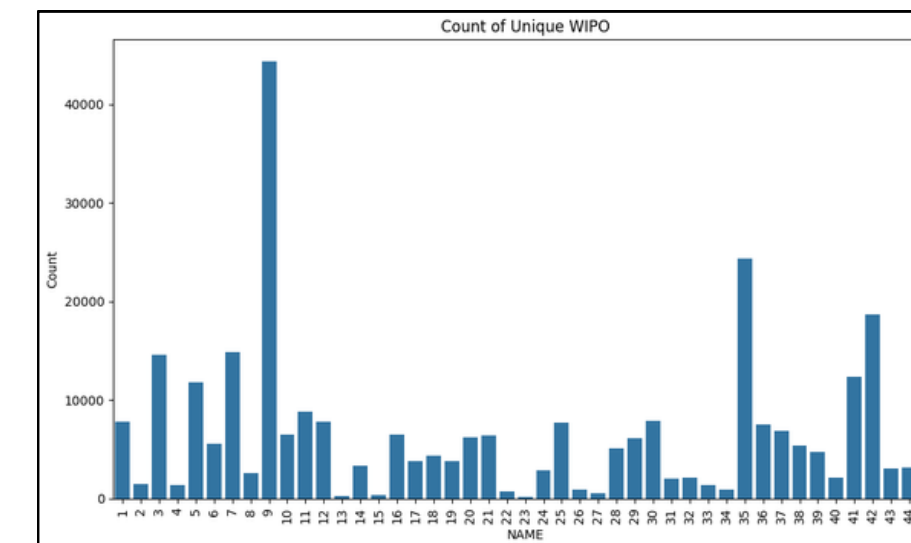
multiple translation



skewness length

ENG	THA
(Animal) skin , pelt	หนัง หนังสัตว์
(audio-video) disc	จานวิดีโอเสียง
(child) car seat	ที่นั่งในรถยนต์สำหรับเด็ก

mismatch punctuation



class imbalance

1. Multiple Translation

~47k rows ENG with multiple THA translation

Solution: LLM as a judge

- gemma3-27b-it
- qwen3-32b-it (wangchan dataset)

	ENG	THA	qwen3
31	Abalone (not live)	หอยเป่าฮื้อ(ไม่มีชีวิต)	หอยเป่าฮื้อ (ไม่ใช่สิ่งมีชีวิต)
32	Abrading instrument [hand instrument]	เครื่องมือขัด [เครื่องมือที่ควบคุมด้วยมือ]	เครื่องมือขัด [เครื่องมือขนาดเล็กทำงานได้ด้วยมือ]
33	Abrasive paper	กระดาษทราย	กระดาษทรายสำหรับการขัด
36	Acanthopanax wine (ogapiju)	ไวน์อะแคนโทพาแนกซ์ (โอกาปิจู)	ไวน์อะแคนโทพาแนกซ์(โอกาปิจู)
37	Acceptance of deposit [including substitute b...	การรับฝากเงิน [รวมถึงการออกพันธบัตรทดแทน] และก...	การรับฝากเงิน [รวมถึงการออกพันธบัตรทดแทน] และก...
38	Accepting , processing , and reconciling finan...	การยอมรับ การประมวลผล และการกระทขยอรายการทางก...	การยอมรับ ประมวลผล และการพิสูจน์ยอดและธุรกรรมก...
39	Accident insurance underwriting	การรับประกันความเสียหายอุบัติเหตุ	การรับประกันภัยอุบัติเหตุ
40	Accommodation bureau (hotel , boarding house)	สำนักงานบริการที่พัก (โรงแรม, บ้านพัก)	สำนักงานให้บริการที่พัก (โรงแรม และบ้านพัก)
41	Accommodation bureau [hotel , boarding house]	สำนักงานจัดหาที่พัก[โรงแรม, หอพัก]	สำนักงานที่พัก [โรงแรม, บ้านพักอาศัย]
42	Accommodation bureau service [hotel , boardin...	บริการสำนักงานจัดหาที่พัก[โรงแรม บ้านพัก]	บริการสำนักงานจัดหาที่พัก[โรงแรม บ้านพักบ้านพั...

Correct format rows

Gemma3: 35,602

Qwen3: 34,913

2. Mismatch punctuation

- gemma3-27b-it

ENG	THA
(Animal) skin , pelt	หนัง หนังสัตว์
(IaaS) infrastructure a a service	ให้บริการโครงสร้างพื้นฐานด้านไอที (ไอเอเอเอส)
(abrasive preparation) soap	(สารที่เตรียมขึ้นใช้ขัด) สบู่
all good of textile	สินค้าทั้งหมดที่ทำจากสิ่งทอ
(audio-video) disc	จานวิดีโอเสียง

ENG	THA
(Animal) skin , pelt	(สัตว์) หนัง , ขนสัตว์
(IaaS) infrastructure a a service	ให้บริการโครงสร้างพื้นฐานด้านไอที (ไอเอเอเอส)
(abrasive preparation) soap	(สารที่เตรียมขึ้นใช้ขัด) สบู่
all good of textile	สินค้าทั้งหมดที่ทำจากสิ่งทอ
(audio-video) disc	(เสียง-วิดีโอ) แผ่นดิสก์

	ENG	THA	NAME
0	(Animal) skin , pelt	(สัตว์) หนัง , ขนสัตว์	['18']
1	(IaaS) infrastructure a a service	ให้บริการโครงสร้างพื้นฐานด้านไอที (ไอเอเอเอส)	['42']
2	(abrasive preparation) soap	(สารที่เตรียมขึ้นใช้ขัด) สบู่	['3']
3	all good of textile	สินค้าทั้งหมดที่ทำจากสิ่งทอ	['24']
4	(audio-video) disc	(เสียง-วิดีโอ) แผ่นดิสก์	['9']
...	...	...	...
200992	zirconia	เซอร์โคเนีย	['1']
200993	zirconium	เซอร์โคเนียม	['6']
200994	zither	ซิทเธอร์ [พิณชนิดหนึ่ง]	['15']
200995	zoological garden service	บริการสวนสัตว์	['41']
200996	zori [Japanese style sandal]	รองเท้าโซริ [รองเท้าแตะแบบญี่ปุ่น]	['25']

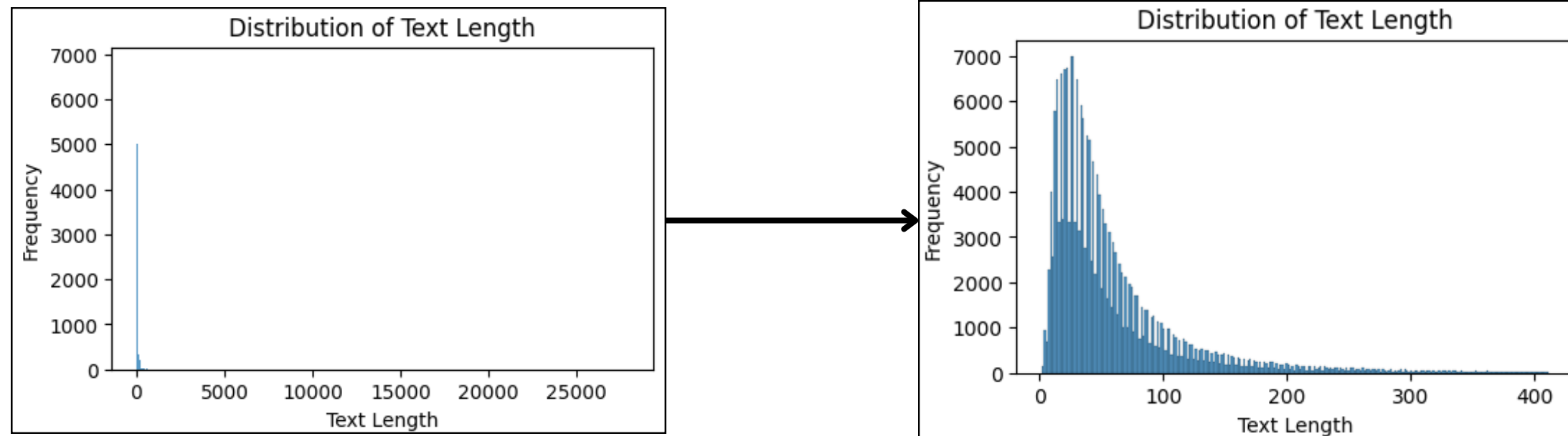
200997 rows × 3 columns

cleaned Result
(~200k rows)

3. Skewness Length

Solution: chunking

- filter text with length $>$ 99th percentile (len=412)
- chunk by punctuation (comma, slash)



Text Normalization

- lemmatize

ENG	ENG
repair of materials and sports equipment	repair of material and sport equipment
electric coffee machines	electric coffee machine
three piece suits [clothing]	three piece suit [clothing]
computer software for obtaining, arranging and...	computer software for obtaining , arranging an...
services related to exploitation and productio...	service related to exploitation and production...

- clean out fullstop

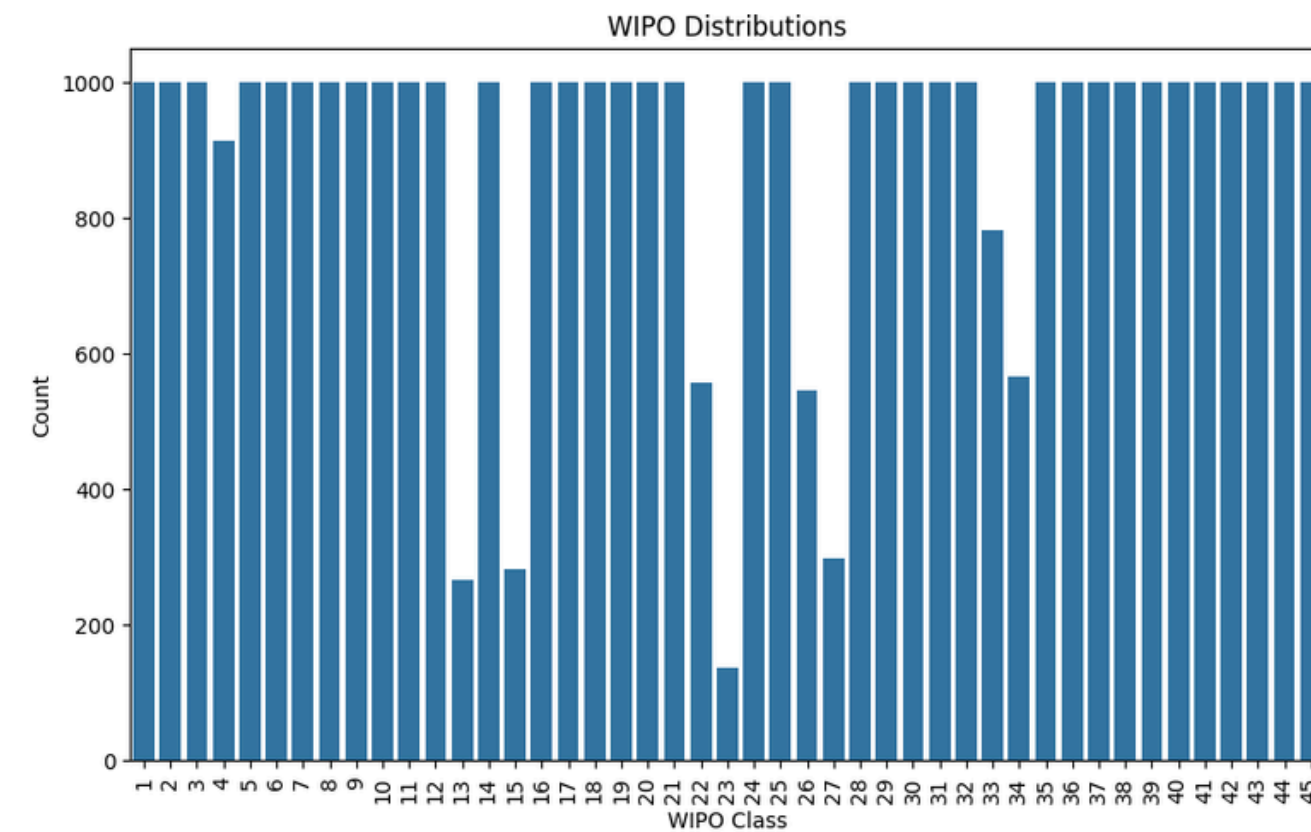
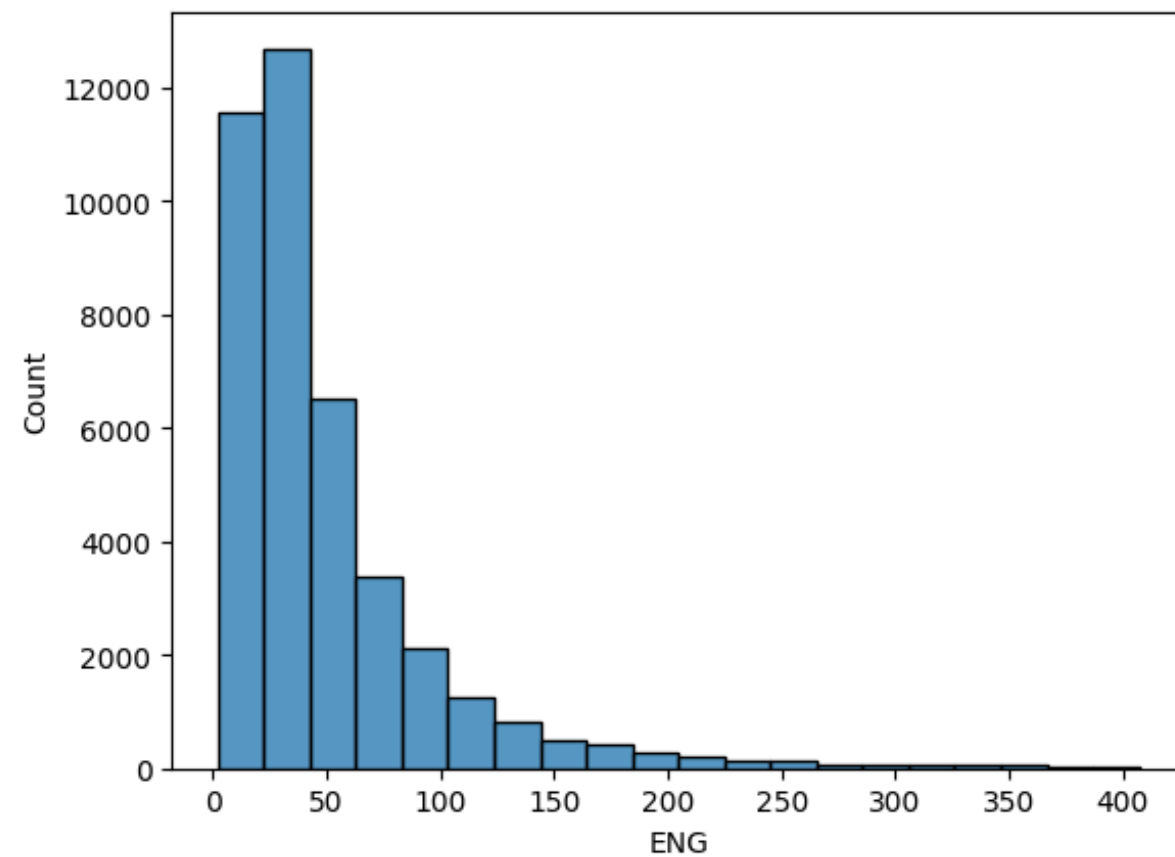
ENG	THA
rum.	สุราชนิดทำจากน้ำอ้อย
Gin.	เหล้าจีน
wax.	ขี้ผึ้ง
fur.	ขนสัตว์
tea.	ชา

- unnecessary parenthesis

ENG	THA
(all good of textile)	สินค้าทั้งหมดที่ทำจากสิ่งทอ
(lap robe)	ผ้าคลุมหน้าตัก
(all aforementioned good included in this class)	สินค้าทั้งหมดที่กล่าวมาข้างต้นรวมอยู่ในจำพวกนี้

4. Class imbalance

- sampling 1k records per class
- result: ~40k records



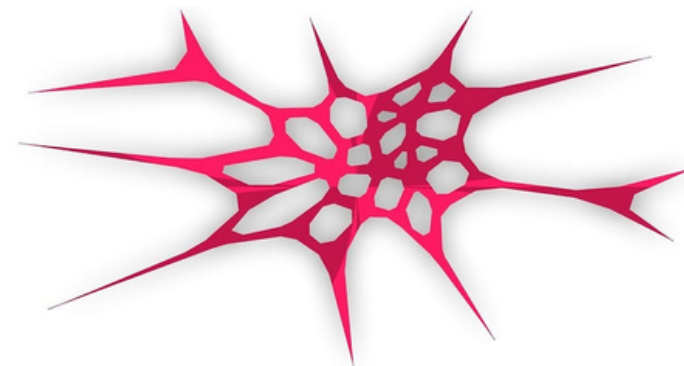
Experiment & Result

Experiment Setup



Fine-tuning

QLoRA rank64



RAG

faiss vectorstore
BGE-m3 retriever



Inference

1 A100 GPU



Evaluation

Attacut tokenizer

Fine-tuning

LoRA: LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS

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ABSTRACT

An important paradigm of natural language processing consists of large-scale pre-training on general domain data and adaptation to particular tasks or domains. As we pre-train larger models, full fine-tuning, which re-trains all model parameters, becomes less feasible. Using GPT-3 175B as an example – deploying independent instances of fine-tuned models, each with 175B parameters, is prohibitively expensive. We propose **Low-Rank Adaptation**, or LoRA, which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture, greatly reducing the number of trainable parameters for downstream tasks. Compared to GPT-3 175B fine-tuned with Adam, LoRA can reduce the number of trainable parameters by 10,000 times and the GPU memory requirement by 3 times. LoRA performs on-par or better than fine-tuning in model quality on RoBERTa, DeBERTa, GPT-2, and GPT-3, despite having fewer trainable parameters, a higher training throughput, and, unlike adapters, *no additional inference latency*. We also provide an empirical investigation into rank-deficiency in language model adaptation, which sheds light on the efficacy of LoRA. We release a package that facilitates the integration of LoRA with PyTorch models and provide our implementations and model checkpoints for RoBERTa, DeBERTa, and GPT-2 at <https://github.com/microsoft/LoRA>.

	Weight Type	$r = 1$	$r = 2$	$r = 4$	$r = 8$	$r = 64$
WikiSQL($\pm 0.5\%$)	W_q	68.8	69.6	70.5	70.4	70.0
	W_q, W_v	73.4	73.3	73.7	73.8	73.5
	W_q, W_k, W_v, W_o	74.1	73.7	74.0	74.0	73.9
MultiNLI ($\pm 0.1\%$)	W_q	90.7	90.9	91.1	90.7	90.7
	W_q, W_v	91.3	91.4	91.3	91.6	91.4
	W_q, W_k, W_v, W_o	91.2	91.7	91.7	91.5	91.4

RAG setup

Retrieval-Augmented Machine Translation with Unstructured Knowledge

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Abstract

Retrieval-augmented generation (RAG) introduces additional information to enhance large language models (LLMs). In machine translation (MT), previous work typically retrieves in-context examples from paired MT corpora, or domain-specific knowledge from knowledge graphs, to enhance MT models. However, a large amount of world knowledge is organized in unstructured documents, and might not be fully paired across different languages. In this paper, we study retrieval-augmented MT using unstructured documents. Specifically, we build RAGtrans, the first benchmark to train and evaluate LLMs' retrieval-augmented MT ability. RAGtrans contains 169K MT samples collected via GPT-4o and human translators. Besides, documents from various languages are also provided to supply the knowledge to these samples. Based on RAGtrans, we further propose a multi-task training method to teach LLMs how to use information from multilingual documents during their translation. The method uses existing multilingual corpora to create auxiliary training objectives without additional labeling requirements. Extensive experiments show that the method improves LLMs by 1.6~3.1 BLEU and 1.0~2.0 COMET scores in En $\Rightarrow$ Zh, and 1.7~2.9 BLEU and 2.1~2.7 COMET scores in En $\Rightarrow$ De. We also conclude the critical difficulties that current LLMs face with this task.¹

1 Introduction

Retrieval-augmented generation (RAG) has grown into a practical paradigm in the development of large language models (LLMs). With the help of retrieved information, LLMs could generate more accurate and knowledge-enrich responses (Li et al., 2022; Gao et al., 2023).

Previous work brought RAG into machine translation (MT), and could be mainly classified into the

following two streams: (1) *Retrieving in-context examples* (also known as “translation memory”): for a source sentence, a few studies retrieve the relevant paired sentences from bilingual corpora to enhance MT models (Zhang et al., 2018; Bulte and Tezcan, 2019; Khandelwal et al., 2021; He et al., 2021; Hoang et al., 2023). Further, Cai et al. (2021) relax the bilingualism limitation, and try to directly retrieve similar target-language translations to enhance models. (2) *Retrieving knowledge triplets*: the others retrieve relevant information from knowledge graphs to let the models know domain or cultural knowledge w.r.t. the source sentences (Conia et al., 2024; Chen et al., 2024b). Despite the great success that has been achieved, a large amount of world knowledge is organized in unstructured documents, and might not be fully paired across different languages. This unstructured knowledge is neglected by previous work. For example, Wikipedia serves as an encyclopedia of world knowledge. Most of its information is listed in documents. Besides, for a piece of specific knowledge, Wikipedia does not always provide it in all languages. Though multilingual information of some general knowledge is provided, their content might be differentiated among different languages (Perez-Beltrachini and Lapata, 2021).

In this paper, we study retrieval-augmented MT using unstructured documents. Since we are the first to study this topic and previous datasets do not support the research, we first build a benchmark dataset, named **RAGtrans**. In detail, RAGtrans is collected based on Wikipedia with three key features: (i) *Knowledge-intensive sentences*: RAGtrans randomly selects 169K English sentences from Wikipedia as the source sentences, which generally come from the lead paragraphs of different Wikipedia pages, containing knowledge-intensive semantics. Thus, understanding these source sentences tends to require additional knowledge. (ii) *Useful relevant documents*: To achieve retrieval-

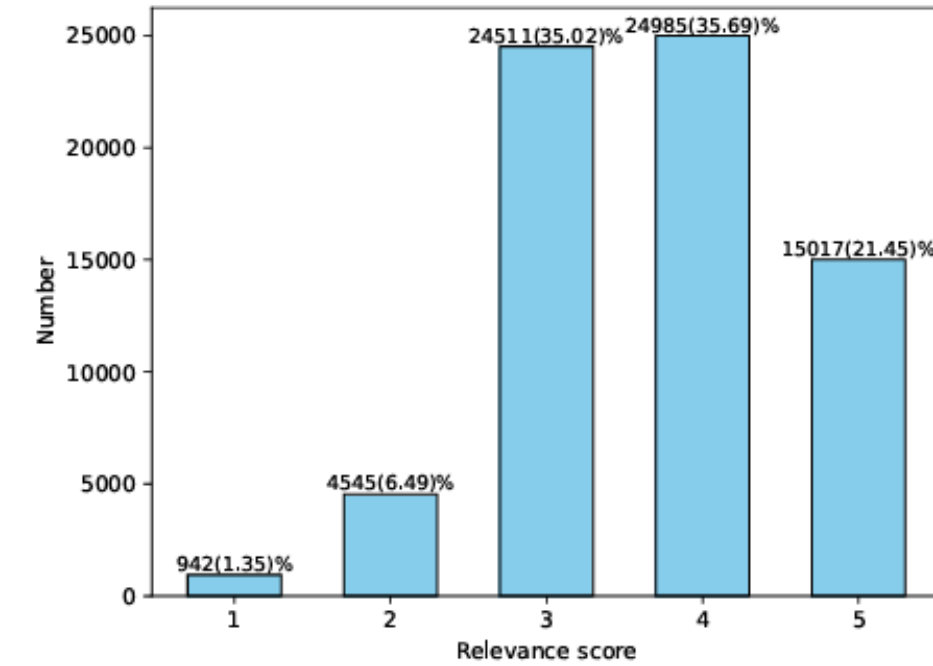


Figure 2: The distribution of relevance scores.

* Corresponding author.

¹<https://github.com/krystalan/RAGtrans>

Base prompt

Instructions:

You are an expert in the classification of goods and services under the WIPO Nice Classification system.
Your task is to translate product names from English to accurate and direct Thai.

****Translation Guidelines:****

- Maintain the punctuation alignment of the input text.
- Use Thai legal and commercial terminology appropriate for trademarks and product classification.
- Do not include explanations, commentary, or any information beyond the translation output.
- Provided output in the format of {"thai_translation": "Thai translation"}

****Product Domain:****

{WIPO}

{RAG_DOC}

Input:

{ENGLISH}

Output:

Benchmark

Model	Metric			Method
	CER	BLEU	COMET	
Qwen3-32B	0.44	0.26	0.87	FT
Gemma3-27B-it	0.39	0.34	0.91	FT
Qwen3-32B	0.40	0.31	0.89	RAG
Gemma3-27B-it	0.36	0.37	0.90	RAG
Qwen3-4B	0.31	0.41	0.90	FFT
Gemma3-4B-pt	0.31	0.45	0.90	FFT
NLLB200-3.3B (Baseline)	0.32	0.46	0.89	FT

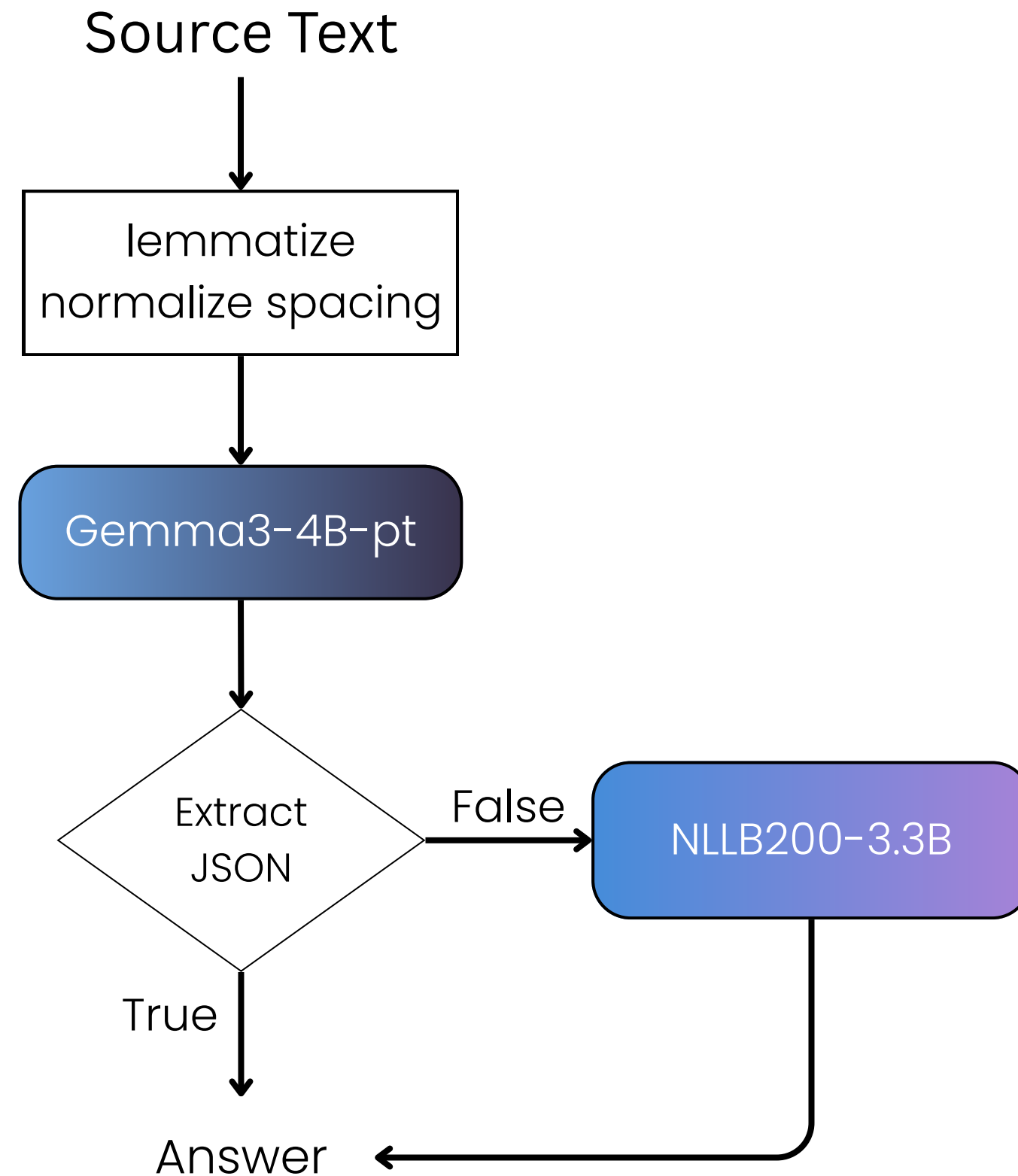
Evaluation

Metric	NLLB200-3.3B	Gemma3-4B-pt (FFT)
Model Size (GB)	6.4	4.3
Peak VRAM Usage (GB)	29	20
Inference Speed (ms/sample)	93	10

Performance

33% reduces in VRAM usage

9.3x faster in inference speed



Future Plan

- deployment and monitoring
- RAG with full-finetuning
- Synthetic data for less WIPO categories

SEASON

5

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